Week-4 (DAY-21) Hands-On. Keerthivasan R

Please download and refer shared Code template (LinqCodeTemplate) and solve problems as given below.

(Please Refer Problem-1 Solved in the Code Template and solve rest of other problems in the same project accordingly)

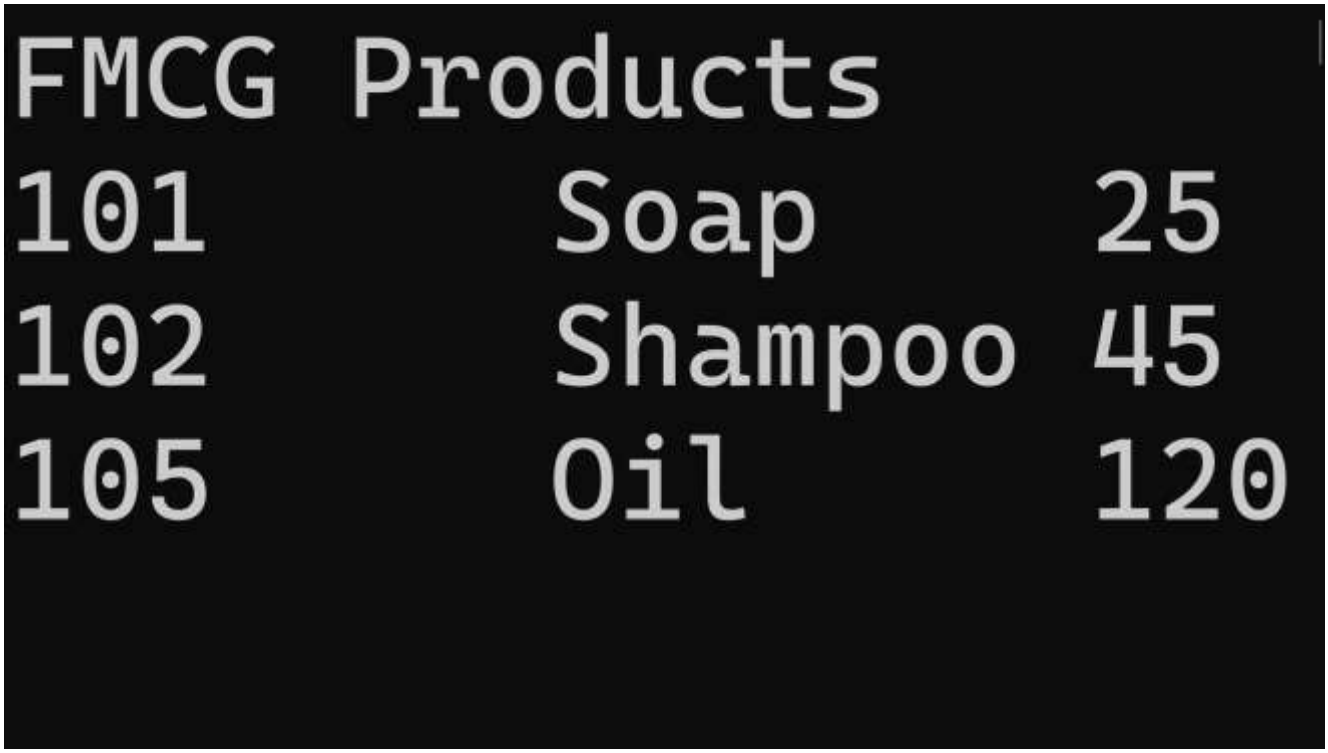
Problem Level- 1 and 2:

1.Write a LINQ query to search and display all products with category "FMCG".

CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace HandsOnDay21
{
    public class Product1
    {
        public int ProCode { get; set; }
        public string ProName { get; set; }
        public string ProCategory { get; set; }
        public double ProMrp { get; set; }

        public List<Product1> GetProducts()
        {
            return new List<Product1>
            {
                new Product1{ ProCode=101, ProName="Soap", ProCategory="FMCG", ProMrp=25
},
                new Product1{ ProCode=102, ProName="Shampoo", ProCategory="FMCG",
ProMrp=45 },
                new Product1{ ProCode=103, ProName="Rice", ProCategory="Grain",
ProMrp=60 },
                new Product1{ ProCode=104, ProName="Wheat", ProCategory="Grain",
ProMrp=40 },
                new Product1{ ProCode=105, ProName="Oil", ProCategory="FMCG", ProMrp=120
}
            };
        }
    }
    internal class LINQ_Query_Problem_01
    {
        static void Main(string[] args) {

            Product1 product = new Product1();

            var products = product.GetProducts();

            var result = products.Where(p => p.ProCategory == "FMCG").ToList();

            foreach (var item in result) {

                Console.WriteLine($"{item.ProCode}\t{item.ProName}\t{item.ProMrp}");
            }
            Console.ReadLine();
        }
    }
}
```

OUTPUT:



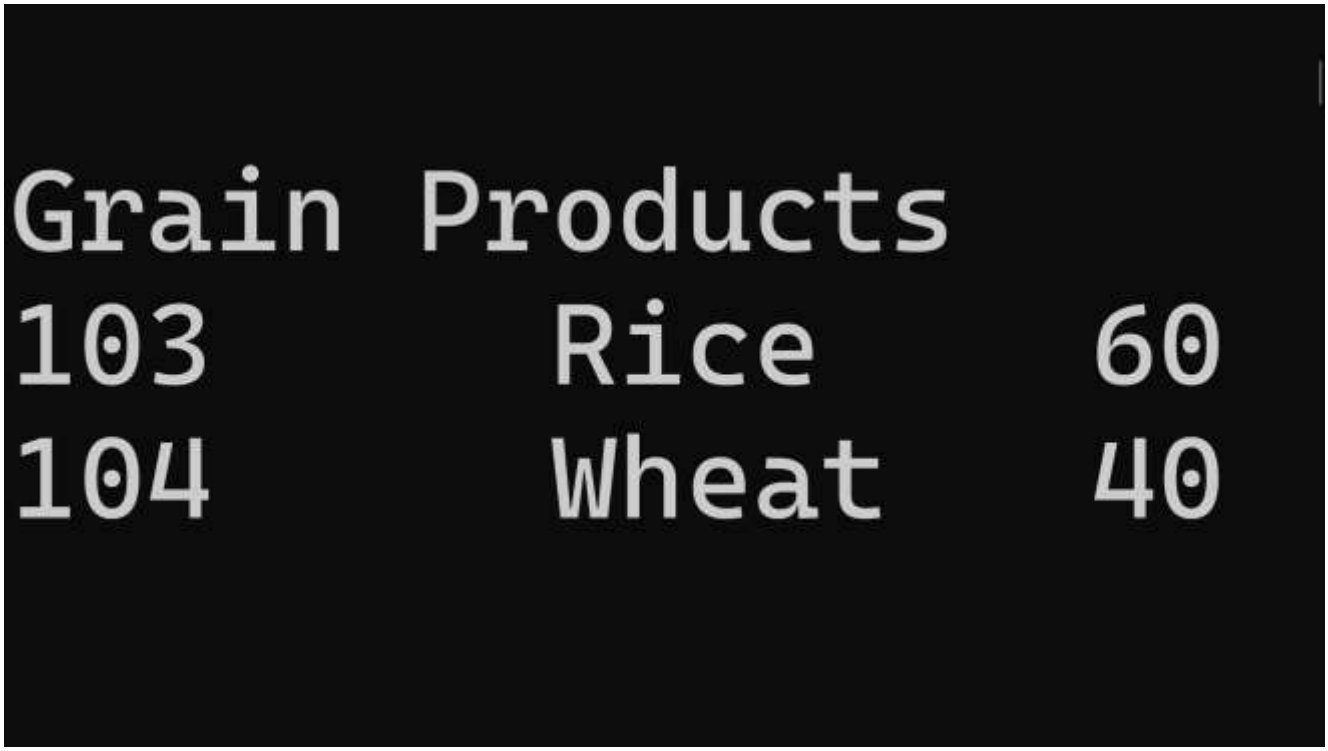
2. Write a LINQ query to search and display all products with category “Grain”.

CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace HandsOnDay21
{
    public class Product2
    {
        public int ProCode { get; set; }
        public string ProName { get; set; }
        public string ProCategory { get; set; }
        public double ProMrp { get; set; }

        public List<Product2> GetProducts()
        {
            return new List<Product2>
            {
                new Product2{ ProCode=101, ProName="Soap", ProCategory="FMCG", ProMrp=25
},
                new Product2{ ProCode=102, ProName="Shampoo", ProCategory="FMCG",
ProMrp=45 },
                new Product2{ ProCode=103, ProName="Rice", ProCategory="Grain",
ProMrp=60 },
                new Product2{ ProCode=104, ProName="Wheat", ProCategory="Grain",
ProMrp=40 },
                new Product2{ ProCode=105, ProName="Oil", ProCategory="FMCG", ProMrp=120
}
            };
        }
    }
    internal class LINQ_Query_Problem_02
    {
```

```
        static void Main(string[] args) {

            Product2 product2 = new Product2();

            //products with category GRAIN
            var products = product2.GetProducts();

            var result = products.Where(p=> p.ProCategory=="Grain").ToList();

            foreach (var item in result) {
                Console.WriteLine($"{item.ProCode}\t{item.ProName}\t{item.ProMrp}");
            }

            Console.ReadLine();

        }
    }
}
```

OUTPUT:



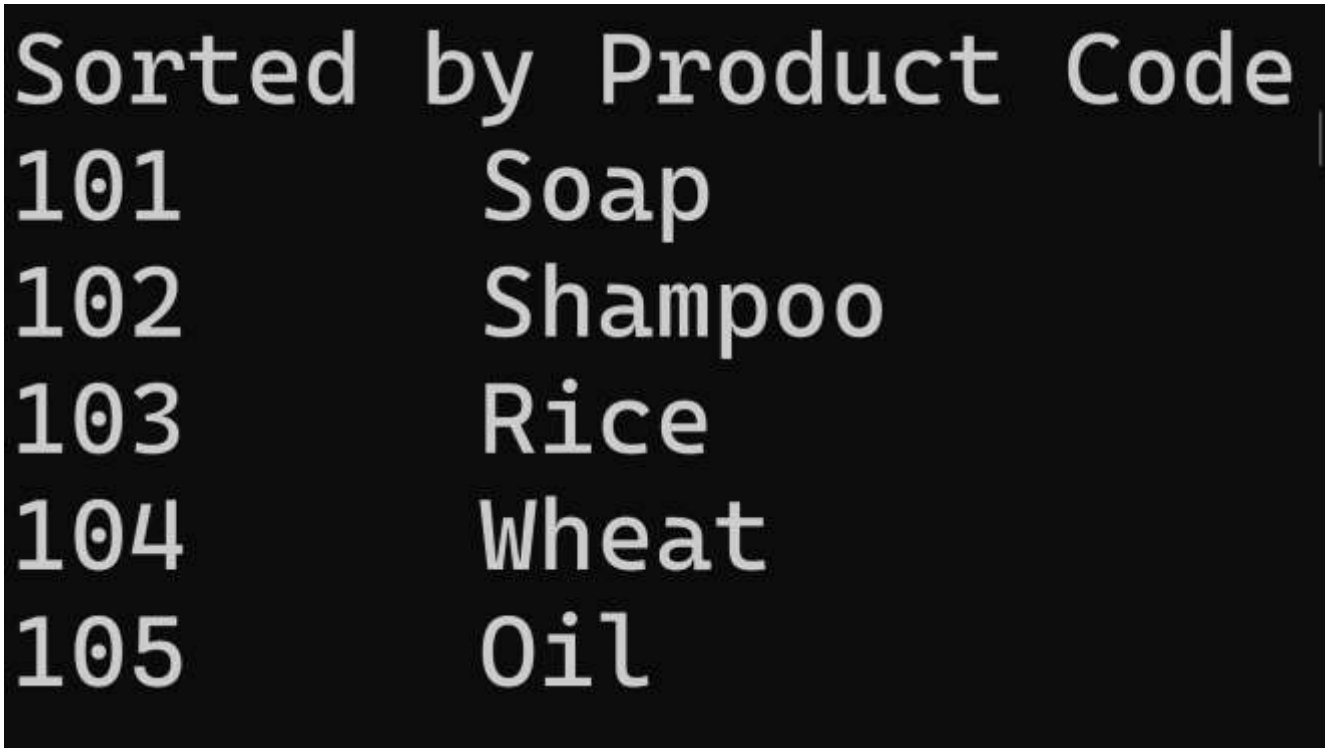
3. Write a LINQ query to sort products in ascending order by product code.

CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace HandsOnDay21
{
    public class Product3
    {
        public int ProCode { get; set; }
        public string ProName { get; set; }
        public string ProCategory { get; set; }
        public double ProMrp { get; set; }

        public List<Product3> GetProducts()
```

```
        {
            return new List<Product3>
            {
                new Product3{ ProCode=101, ProName="Soap", ProCategory="FMCG", ProMrp=25
},
                new Product3{ ProCode=102, ProName="Shampoo", ProCategory="FMCG",
ProMrp=45 },
                new Product3{ ProCode=103, ProName="Rice", ProCategory="Grain",
ProMrp=60 },
                new Product3{ ProCode=104, ProName="Wheat", ProCategory="Grain",
ProMrp=40 },
                new Product3{ ProCode=105, ProName="Oil", ProCategory="FMCG", ProMrp=120
}
            };
        }
    }
    internal class LINQ_Query_Problem_03
    {
        static void Main(string[] args) {

            Product3 product = new Product3();

            // 1. Get the list from the method
            var productList = product.GetProducts();

            // 2. Sort the list, not the single 'product' object
            var result3 = productList.OrderBy(p => p.ProCode);

            Console.WriteLine("\nSorted by Product Code:");
            foreach (var item in result3)
                Console.WriteLine($"{item.ProCode}\t{item.ProName}");
        }
    }
}
```

OUTPUT:

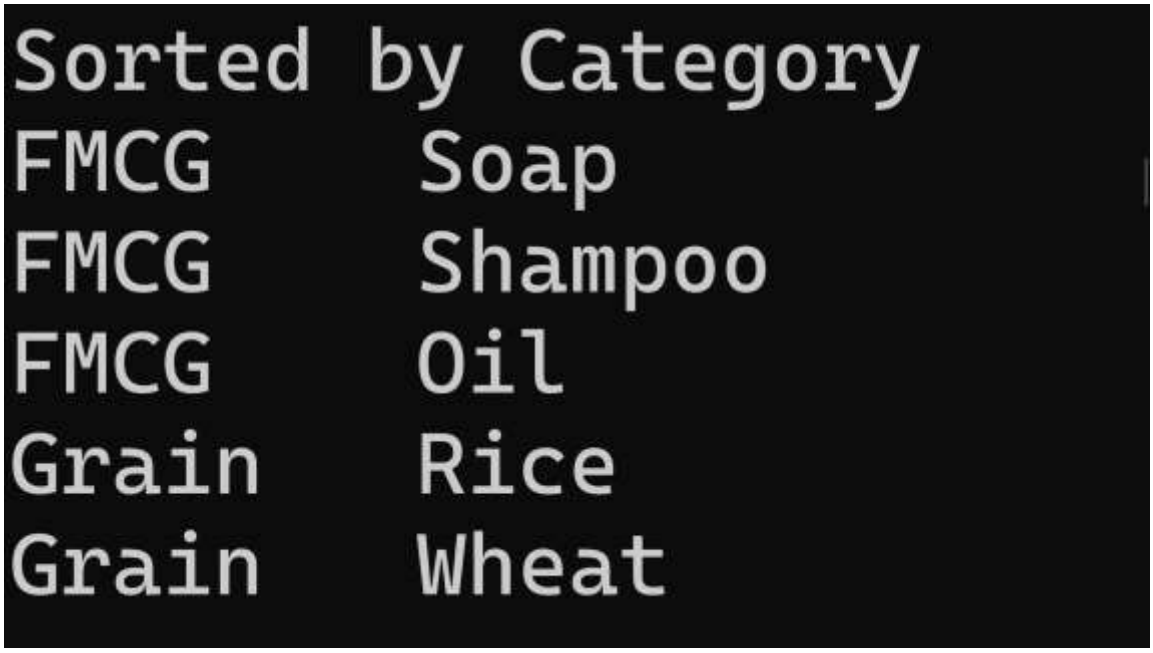


4. Write a LINQ query to sort products in ascending order by product Category.

CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace HandsOnDay21
{
    public class Product4
    {
        public int ProCode { get; set; }
        public string ProName { get; set; }
        public string ProCategory { get; set; }
        public double ProMrp { get; set; }
        public List<Product4> GetProducts()
        {
            return new List<Product4>
            {
                new Product4{ ProCode=101, ProName="Soap", ProCategory="FMCG", ProMrp=25
},
                new Product4{ ProCode=102, ProName="Shampoo", ProCategory="FMCG",
ProMrp=45 },
                new Product4{ ProCode=103, ProName="Rice", ProCategory="Grain",
ProMrp=60 },
                new Product4{ ProCode=104, ProName="Wheat", ProCategory="Grain",
ProMrp=40 },
                new Product4{ ProCode=105, ProName="Oil", ProCategory="FMCG", ProMrp=120
}
            };
        }
    }
    internal class LINQ_Query_Problem_04
    {
        static void Main(string[] args) {
            Product4 product = new Product4();
            //sort products in ascending order by product Category
            var products = product.GetProducts();
            var result = products.OrderBy(p => p.ProCategory);
            foreach (var item in result) {
                Console.WriteLine($"{item.ProCategory}\t{item.ProName}");
            }
            Console.ReadLine();
        }
    }
}
```

OUTPUT:

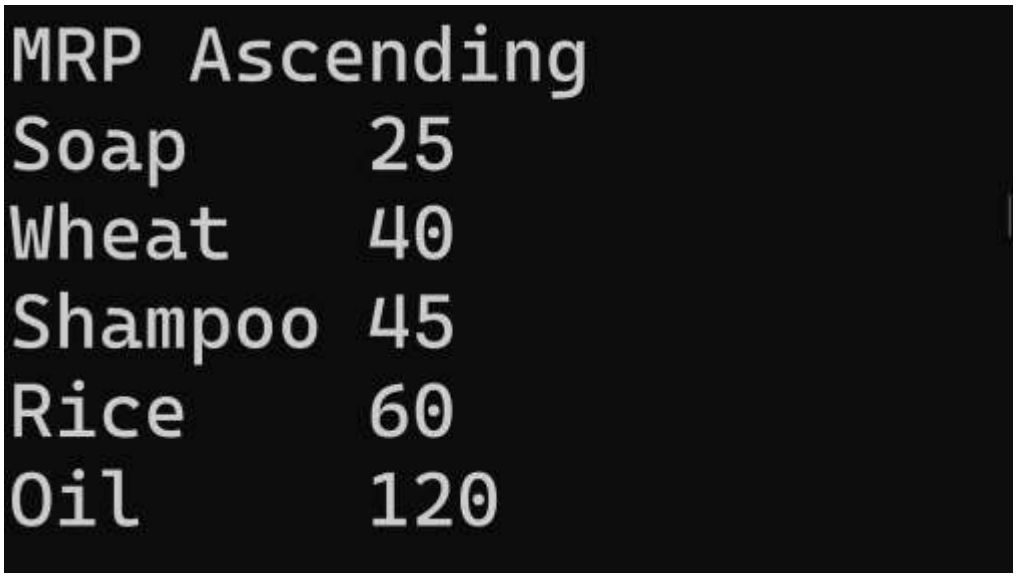


5. Write a LINQ query to sort products in ascending order by product Mrp.

CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace HandsOnDay21
{
    public class Product5
    {
        public int ProCode { get; set; }
        public string ProName { get; set; }
        public string ProCategory { get; set; }
        public double ProMrp { get; set; }
        public List<Product5> GetProducts()
        {
            return new List<Product5>
            {
                new Product5{ ProCode=101, ProName="Soap", ProCategory="FMCG", ProMrp=25
},
                new Product5{ ProCode=102, ProName="Shampoo", ProCategory="FMCG",
ProMrp=45 },
                new Product5{ ProCode=103, ProName="Rice", ProCategory="Grain",
ProMrp=60 },
                new Product5{ ProCode=104, ProName="Wheat", ProCategory="Grain",
ProMrp=40 },
                new Product5{ ProCode=105, ProName="Oil", ProCategory="FMCG", ProMrp=120
}
            };
        }
    }
    internal class LINQ_Query_Problem_05
    {
        static void Main(string[] args) {
            Product5 product = new Product5();
            //sort products in ascending order by product Mrp
            var products = product.GetProducts();
            var result = products.OrderBy(p=> p.ProMrp).ToList();
            foreach (var item in result)
            {
                Console.WriteLine($"{item.ProMrp}\t{item.ProName}");
            }
            Console.ReadLine();
        }
    }
}
```

OUTPUT:

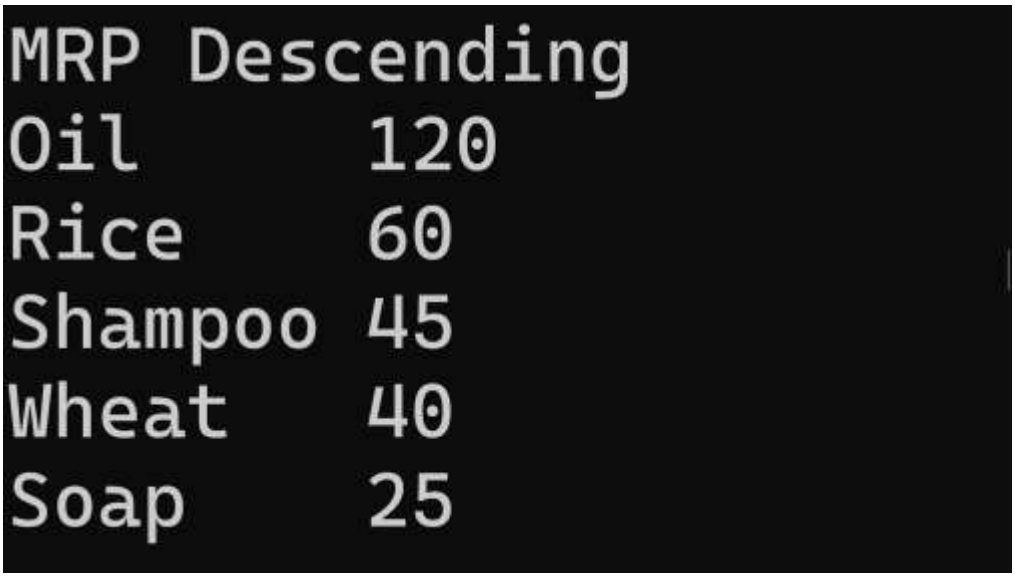


6. Write a LINQ query to sort products in descending order by product Mrp.

CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace HandsOnDay21
{
    public class Product6
    {
        public int ProCode { get; set; }
        public string ProName { get; set; }
        public string ProCategory { get; set; }
        public double ProMrp { get; set; }
        public List<Product6> GetProducts()
        {
            return new List<Product6>
            {
                new Product6{ ProCode=101, ProName="Soap", ProCategory="FMCG", ProMrp=25
},
                new Product6{ ProCode=102, ProName="Shampoo", ProCategory="FMCG",
ProMrp=45 },
                new Product6{ ProCode=103, ProName="Rice", ProCategory="Grain",
ProMrp=60 },
                new Product6{ ProCode=104, ProName="Wheat", ProCategory="Grain",
ProMrp=40 },
                new Product6{ ProCode=105, ProName="Oil", ProCategory="FMCG", ProMrp=120
}
            };
        }
    }
    internal class LINQ_Query_Problem_06
    {
        static void Main(string[] args) {
            Product6 product6 = new Product6();
            //sort products in descending order by product Mrp
            var products = product6.GetProducts();
            var result = products.OrderByDescending(p=> p.ProMrp).ToList();
            foreach (var item in products)
            {
                Console.WriteLine($"{item.ProName}\t{item.ProMrp}");
            }
            Console.ReadLine();
        }
    }
}
```

OUTPUT:

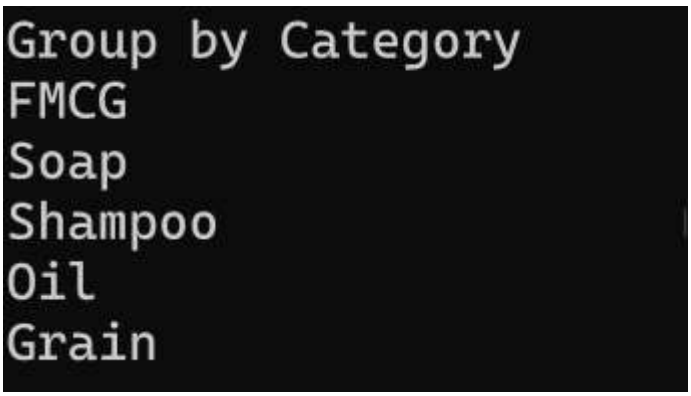


7. Write a LINQ query to display products group by product Category.

CODE:

```
using System;

using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace HandsOnDay21
{
    public class Product7
    {
        public int ProCode { get; set; }
        public string ProName { get; set; }
        public string ProCategory { get; set; }
        public double ProMrp { get; set; }

        public List<Product7> GetProducts()
        {
            return new List<Product7>
            {
                new Product7{ ProCode=101, ProName="Soap", ProCategory="FMCG", ProMrp=25
},
                new Product7{ ProCode=102, ProName="Shampoo", ProCategory="FMCG",
ProMrp=45 },
                new Product7{ ProCode=103, ProName="Rice", ProCategory="Grain",
ProMrp=60 },
                new Product7{ ProCode=104, ProName="Wheat", ProCategory="Grain",
ProMrp=40 },
                new Product7{ ProCode=105, ProName="Oil", ProCategory="FMCG", ProMrp=120
}
            };
        }
    }
    internal class LINQ_Query_Problem_07
    {
        static void Main(string[] args)
        {
            Product7 product7 = new Product7();
            var products = product7.GetProducts();
            // Perform the grouping
            var result = products.GroupBy(p => p.ProCategory).ToList();
            // Iterate over 'result', NOT 'products'
            foreach (var group in result)
            {
                // group.Key is the Category Name (e.g., "FMCG" or "Grain")
                Console.WriteLine($"\nCategory: {group.Key}");
                // Each 'group' acts like a list of products within that category
                foreach (var item in group)
                {
                    Console.WriteLine($" - {item.ProName}: {item.ProMrp}");
                }
            }
            Console.ReadLine();
        }
    }
}
```

OUTPUT:



8. Write a LINQ query to display products group by product Mrp.

CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace HandsOnDay21
{
    public class Product8    {
        public int ProCode { get; set; }
        public string ProName { get; set; }
        public string ProCategory { get; set; }
        public double ProMrp { get; set; }

        public List<Product8> GetProducts()        {
            return new List<Product8>            {
                new Product8{ ProCode=101, ProName="Soap", ProCategory="FMCG", ProMrp=25
},
                new Product8{ ProCode=102, ProName="Shampoo", ProCategory="FMCG",
ProMrp=45 },
                new Product8{ ProCode=103, ProName="Rice", ProCategory="Grain",
ProMrp=60 },
                new Product8{ ProCode=104, ProName="Wheat", ProCategory="Grain",
ProMrp=40 },
                new Product8{ ProCode=105, ProName="Oil", ProCategory="FMCG", ProMrp=120
}
            };
        }
    }
    internal class LINQ_Query_Problem_08
    {
        static void Main(string[] args) {
            Product8 product = new Product8();
            var products = product.GetProducts();
            // 1. Group the products by Mrp
            var result = products.GroupBy(p => p.ProMrp).ToList();
            Console.WriteLine("\nGroup by MRP:");
            // 2. Loop through 'result' (the groups), NOT 'products' (the raw list)
            foreach (var group in result)            {
                // group.Key is the actual MRP value (e.g., 25, 45, 60...)
                Console.WriteLine("\nMRP: " + group.Key);
                // Loop through all products that have this specific MRP
                foreach (var item in group)
                {
                    Console.WriteLine($" - {item.ProName}");
                }
            }
        }
    }

}
```

OUTPUT:

9. Write a LINQ query to display product detail with highest price in FMCG category.

CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace HandsOnDay21
    public class Product9    {
        public int ProCode { get; set; }
        public string ProName { get; set; }
        public string ProCategory { get; set; }
        public double ProMrp { get; set; }

        public List<Product9> GetProducts()        {
            return new List<Product9>
            {
                new Product9{ ProCode=101, ProName="Soap", ProCategory="FMCG", ProMrp=25
},
                new Product9{ ProCode=102, ProName="Shampoo", ProCategory="FMCG",
ProMrp=45 },
                new Product9{ ProCode=103, ProName="Rice", ProCategory="Grain",
ProMrp=60 },
                new Product9{ ProCode=104, ProName="Wheat", ProCategory="Grain",
ProMrp=40 },
                new Product9{ ProCode=105, ProName="Oil", ProCategory="FMCG", ProMrp=120
}
            };        }    }
    internal class LINQ_Query_Problem_09    {
        static void Main(string[] args) {
            Product9 product9 = new Product9();
            var products = product9.GetProducts();
            // 1. Query 'products' (the List), not 'product9'
            // 2. Remove the extra dot at the end
            var result9 = products
                .Where(p => p.ProCategory == "FMCG")
                .OrderByDescending(p => p.ProMrp)
                .FirstOrDefault();
            Console.WriteLine("\nHighest FMCG Product:");
            // 3. Simple null check to prevent a NullReferenceException
            if (result9 != null)
            {
                Console.WriteLine($"{result9.ProName} {result9.ProMrp}");
            }
            else
            {
                Console.WriteLine("No FMCG products found.");
            }        }    }
}
```

OUTPUT:

10. Write a LINQ query to display count of total products.

CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
```

```
using System.Text;
using System.Threading.Tasks;
namespace HandsOnDay21{
    public class Product10    {
        public int ProCode { get; set; }
        public string ProName { get; set; }
        public string ProCategory { get; set; }
        public double ProMrp { get; set; }
        public List<Product10> GetProducts()        {
            return new List<Product10>            {
                new Product10{ ProCode=101, ProName="Soap", ProCategory="FMCG",
ProMrp=25 },
                new Product10{ ProCode=102, ProName="Shampoo", ProCategory="FMCG",
ProMrp=45 },
                new Product10{ ProCode=103, ProName="Rice", ProCategory="Grain",
ProMrp=60 },
                new Product10{ ProCode=104, ProName="Wheat", ProCategory="Grain",
ProMrp=40 },
                new Product10{ ProCode=105, ProName="Oil", ProCategory="FMCG",
ProMrp=120 }
            };        }    }
    internal class LINQ_Query_Problem_10    {
        static void Main(string[] args) {
            Product10 product = new Product10();
            var products = product.GetProducts();
            var result10 = products.Count();
            Console.WriteLine("\nTotal Products: " + result10);
        }    }}
```

OUTPUT:

11. Write a LINQ query to display count of total products with category FMCG.

CODE:

```
using System;
using System.Collections.Generic;
using System.Data;
using System.Linq;
using System.Runtime.InteropServices;
using System.Text;
using System.Threading.Tasks;

namespace HandsOnDay21
{
    public class Product11
    {
        public int ProCode { get; set; }
        public string ProName { get; set; }
        public string ProCategory { get; set; }
        public double ProMrp { get; set; }

        public List<Product11> GetProducts()
        {
            return new List<Product11>
            {
                new Product11{ ProCode=101, ProName="Soap", ProCategory="FMCG",
ProMrp=25 },
                new Product11{ ProCode=102, ProName="Shampoo", ProCategory="FMCG",
ProMrp=45 },
```

```
                new Product11{ ProCode=103, ProName="Rice", ProCategory="Grain",
ProMrp=60 },
                new Product11{ ProCode=104, ProName="Wheat", ProCategory="Grain",
ProMrp=40 },
                new Product11{ ProCode=105, ProName="Oil", ProCategory="FMCG",
ProMrp=120 }
            };
        }
    }
    internal class LINQ_Query_Problem_11
    {
        static void Main(string[] args) {

            Product11 product = new Product11();

            var products = product.GetProducts();

            var result11 = products.Count(p => p.ProCategory == "FMCG");
            Console.WriteLine("FMCG Product Count: " + result11);
        }
    }
}
```

OUTPUT:

12.Write a LINQ query to display Max price.

CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace HandsOnDay21
{
    public class Product12
    {
        public int ProCode { get; set; }
        public string ProName { get; set; }
        public string ProCategory { get; set; }
        public double ProMrp { get; set; }

        public List<Product12> GetProducts()
        {
            return new List<Product12>
            {
                new Product12{ ProCode=101, ProName="Soap", ProCategory="FMCG",
ProMrp=25 },
                new Product12{ ProCode=102, ProName="Shampoo", ProCategory="FMCG",
ProMrp=45 },
                new Product12{ ProCode=103, ProName="Rice", ProCategory="Grain",
ProMrp=60 },
                new Product12{ ProCode=104, ProName="Wheat", ProCategory="Grain",
ProMrp=40 },
                new Product12{ ProCode=105, ProName="Oil", ProCategory="FMCG",
ProMrp=120 }
```

```
            };
        }
    }
    internal class LINQ_Query_Problem_12
    {
        static void Main(string[] args) {

            Product12 product = new Product12();

            var products = product.GetProducts();

            var result12 = products.Max(p => p.ProMrp);
            Console.WriteLine("Max Price: " + result12);
        }
    }
}
```

OUTPUT:

```
Max Price: 120
```

13.Write a LINQ query to display Min price.

CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace HandsOnDay21
{
    public class Product13
    {
        public int ProCode { get; set; }
        public string ProName { get; set; }
        public string ProCategory { get; set; }
        public double ProMrp { get; set; }

        public List<Product13> GetProducts()
        {
            return new List<Product13>
            {
                new Product13{ ProCode=101, ProName="Soap", ProCategory="FMCG",
ProMrp=25 },
                new Product13{ ProCode=102, ProName="Shampoo", ProCategory="FMCG",
ProMrp=45 },
                new Product13{ ProCode=103, ProName="Rice", ProCategory="Grain",
ProMrp=60 },
                new Product13{ ProCode=104, ProName="Wheat", ProCategory="Grain",
ProMrp=40 },
                new Product13{ ProCode=105, ProName="Oil", ProCategory="FMCG",
ProMrp=120 }
            };
        }
    }
    internal class LINQ_Query_Problem_13
    {
        static void Main(string[] args) {

            Product13 product = new Product13();

            var products = product.GetProducts();
```

```
            var result13 = products.Min(p => p.ProMrp);
            Console.WriteLine("Min Price: " + result13);
        }
    }
}
```

OUTPUT:

14. Write a LINQ query to display whether all products are below Mrp Rs.30 or not.

CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace HandsOnDay21
{
    public class Product14
    {
        public int ProCode { get; set; }
        public string ProName { get; set; }
        public string ProCategory { get; set; }
        public double ProMrp { get; set; }

        public List<Product14> GetProducts()
        {
            return new List<Product14>
            {
                new Product14{ ProCode=101, ProName="Soap", ProCategory="FMCG",
ProMrp=25 },
                new Product14{ ProCode=102, ProName="Shampoo", ProCategory="FMCG",
ProMrp=45 },
                new Product14{ ProCode=103, ProName="Rice", ProCategory="Grain",
ProMrp=60 },
                new Product14{ ProCode=104, ProName="Wheat", ProCategory="Grain",
ProMrp=40 },
                new Product14{ ProCode=105, ProName="Oil", ProCategory="FMCG",
ProMrp=120 }
            };
        }
    }
    internal class LINQ_Query_Problem_14
    {
        static void Main(string[] args) {

            Product14 product = new Product14();

            var products = product.GetProducts();

            var result14 = products.All(p => p.ProMrp < 30);
            Console.WriteLine("All products below 30: " + result14);
        }
    }
```

}

OUTPUT:

15. Write a LINQ query to display whether any products are below Mrp Rs.30 or not.

CODE:

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
namespace HandsOnDay21
{
    public class Product15
    {
        public int ProCode { get; set; }
        public string ProName { get; set; }
        public string ProCategory { get; set; }
        public double ProMrp { get; set; }
        public List<Product15> GetProducts()
        {
            return new List<Product15>
            {
                new Product15{ ProCode=101, ProName="Soap", ProCategory="FMCG",
ProMrp=25 },
                new Product15{ ProCode=102, ProName="Shampoo", ProCategory="FMCG",
ProMrp=45 },
                new Product15{ ProCode=103, ProName="Rice", ProCategory="Grain",
ProMrp=60 },
                new Product15{ ProCode=104, ProName="Wheat", ProCategory="Grain",
ProMrp=40 },
                new Product15{ ProCode=105, ProName="Oil", ProCategory="FMCG",
ProMrp=120 }
            };
        }
    }
    internal class LINQ_Query_Problem_15
    {
        static void Main(string[] args) {
            Product15 product =new Product15();
            var products = product.GetProducts();
            var result15 = products.Any(p => p.ProMrp < 30);
            Console.WriteLine("Any product below 30: " + result15);

        }
    }
}
```

OUTPUT: